

# Graduate Research Project Proposal

## Project Overview

|                      |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |
|----------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Brief project title: | Open-Source ESG and LCA Tool and Database with LLM-Enabled AI Toolkit for Australia's Sustainable Industries                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |
| Project Description: | <p>A major barrier to scaling sustainable technologies in Australia is the limited availability of consistent, verified Life Cycle Assessment (LCA) data and greenhouse gas (GHG) emission factors. Current databases are fragmented, proprietary, or poorly adapted to Australian industrial contexts, making technology comparison and policy evaluation difficult. This project will deliver Australia's first open-source LCA and ESG database, coupled with an LLM-enabled AI toolkit to automate data extraction, validation, and integration of emission factors. The system will ingest and standardise data from literature, enterprise reports, and industrial datasets, with a focus on unit process detail, traceability, and interoperability with ILCD and international standards. Core tasks include: (i) construction of an open GHG emission factor library, (ii) AI-assisted filling of data gaps in LCA inventories, (iii) cross-industry benchmarking functions for emerging technologies, and (iv) release of an open-source platform with APIs for researchers, SMEs, and industrial stakeholders.</p> |

## Field of Research

|                          |                                    |
|--------------------------|------------------------------------|
| The first research area  | Engineering                        |
| The second research area | Information and Computing Sciences |
| The third research area  | Environmental Sciences             |

# Graduate Research Project Proposal

|                                                            |                                                   |
|------------------------------------------------------------|---------------------------------------------------|
| The research project college alignment                     | College of Engineering and Information Technology |
| AU Signature Research Theme (SRT)                          | Sustainable Green Transition                      |
| Campus the project will be based at                        | Adelaide City                                     |
| Application close date                                     | 31/10/2025                                        |
| Project available to applicants of the following Degree(s) | Doctor of Philosophy<br>Master of Research        |
| Project available to applicants                            | Domestic and International                        |
| Proposed funding model for this project                    | Project Round - Scholarship Funding               |

## Project Details

| Anticipated Outcomes                                                       |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |
|----------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Anticipated Outcomes of the project, output, engagement and future impact: | <ul style="list-style-type: none"> <li>• Open-source national LCA and ESG database, adapted to Australian industrial sectors.</li> <li>• Comprehensive GHG emission factor library for energy, agriculture, transport, and heavy industry.</li> <li>• AI-enabled toolkit for automated extraction and validation of emission factors and LCA datasets.</li> <li>• Benchmarking functions for cross-technology comparisons, supporting policy, R&amp;D, and industry adoption.</li> <li>• Publicly released platform with APIs and documentation for integration into research and industrial workflows.</li> <li>• High-impact publications and conference outputs in sustainability engineering, AI, and industrial decarbonisation.</li> </ul> |
| Research Internships                                                       |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |

# Graduate Research Project Proposal

|                                                                                                                                                                                      |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |
|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <p>Will the successful applicant have the opportunity to undertake a suitable industry placement:</p>                                                                                | <p>Yes. The successful applicant will collaborate with industrial partners, SMEs, and government agencies working on sustainability reporting and carbon accounting, including placements in data collection, validation, and GHG benchmarking case studies.</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             |
| <p><b>Supervisor and research capacity</b></p>                                                                                                                                       |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |
| <p>Describe the capacity and track record (relative to opportunity) of the supervisory panel and how the project contributes to development of supervisory and research capacity</p> | <p>Dr Nam Nghiep Tran, has internationally recognised expertise in sustainable process design, decarbonisation, green chemistry, computational modelling, sustainability assessment and frameworks. Dr Tran holds an H-index of 25 and an i10-index of 45, reflecting a strong record of impactful research published in leading Q1 journals. The group has contributed extensively to national and international initiatives in carbon capture, industrial decarbonisation, and sustainability assessment, ensuring alignment with both academic excellence and industry priorities. The team has recently received two state-level CSIRO awards for sustainable engineering tool development, alongside international recognition from partners in Vietnam and Northern Europe for advancing sustainable process design. Strong collaborations with industry, including HILT CRC, FarmN, and domestic small to large scale industrial stakeholders. Recent industrial projects have ability to translate research into applied outcomes that directly support Australia's industrial decarbonisation and green transition agenda.</p> <p>This project builds supervisory capacity by embedding multidisciplinary expertise in chemical engineering, process modelling, carbon accounting, and AI-enabled sustainability assessment. It will train HDR students with skills of national importance, equipping the next generation of researchers and practitioners to strengthen Australia's sovereign capability in digital sustainability infrastructure and low-carbon technologies.</p> |

# Graduate Research Project Proposal

|                                                                                                                                                                                                                                                     |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |
|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|                                                                                                                                                                                                                                                     |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |
| <b>Project co-design</b>                                                                                                                                                                                                                            |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |
| What is the research project co-design?                                                                                                                                                                                                             | Yes. The project was developed in response to industry and policy demand for transparent GHG emission factors and accessible LCA databases, particularly for SMEs and sustainability practitioners who face barriers in accessing proprietary datasets.                                                                                                                                                                                                                                             |
| <b>Defence Project</b>                                                                                                                                                                                                                              |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |
| What is the research project defence related?                                                                                                                                                                                                       | No. This project does not involve defence-related research.                                                                                                                                                                                                                                                                                                                                                                                                                                         |
| <b>Aboriginal and Torres Strait Islander peoples research</b>                                                                                                                                                                                       |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |
| Is the project led by Aboriginal and Torres Strait Islander Peoples researchers, partnered with Aboriginal and Torres Strait Islander Peoples communities/ organisations, or of benefit to the Aboriginal and Torres Strait Islander Peoples world? | The project indirectly benefits Aboriginal and Torres Strait Islander Peoples by enabling better data transparency and accountability in industrial emissions reporting, advancing caring for Country principles. By providing open-access sustainability tools, the project creates opportunities for participation by Indigenous researchers and organisations. It aligns with the First Nations Academy by embedding equity and inclusivity into Australia's sustainability data infrastructure. |
| <b>Non-scholarship projects costs</b>                                                                                                                                                                                                               |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |
| What is the expected cost of delivering this project, excluding scholarship costs?                                                                                                                                                                  | AUD 20,000–30,000, covering cloud computing, database hosting, API development, industrial data workshops, and dissemination.                                                                                                                                                                                                                                                                                                                                                                       |
| <b>Source of project costs</b>                                                                                                                                                                                                                      |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |
| Describe how the project costs will be met                                                                                                                                                                                                          | Research funding will be sought through government programs and industry partnerships. Core project costs will                                                                                                                                                                                                                                                                                                                                                                                      |

# Graduate Research Project Proposal

|                                                                                                                                             |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          |
|---------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|                                                                                                                                             | be supported by Adelaide University research schemes, with additional contributions anticipated from agricultural industry engagement and national or state-level sustainability research initiatives.                                                                                                                                                                                                                                                                                                   |
| <b>Additional selection criteria</b>                                                                                                        |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          |
| Provide any specific additional selection criteria you would like the student to address in their application                               | <ul style="list-style-type: none"> <li>• First Class Honours or Master's in Data Science, Computer Science, Process Systems, or Chemical Engineering.</li> <li>• Proficiency in coding (Python, R, or equivalent).</li> <li>• Knowledge of LCA methodology and GHG emission factor compilation.</li> </ul> <p>Strong technical writing and analytical skills.</p> <ul style="list-style-type: none"> <li>• For international applicants: IELTS 6.5 overall, no band &lt; 6.0 (or equivalent).</li> </ul> |
| <b>Preferred Applicant</b>                                                                                                                  |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          |
| Have you identified a preferred applicant/group for applicants for this project? They must have a first class Honours/Masters or equivalent | <ul style="list-style-type: none"> <li>• Experience with LCA software and databases (e.g., ecoinvent, ILCD).</li> <li>• Familiarity with carbon accounting and ESG reporting frameworks.</li> <li>• Skills in AI/LLM model application for data extraction and validation.</li> <li>• Ability to work across industrial datasets from multiple sectors (energy, mining, agriculture, manufacturing).</li> </ul>                                                                                          |
| Is the project available on a part-time basis?                                                                                              | No                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |
| If the project is not successful in the next project round, I would like the project to be advertised on the web                            | Yes                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      |

# Graduate Research Project Proposal

## Principal Supervisor

|                                                                      |                                                                                                                                     |
|----------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------|
| Title                                                                | Dr                                                                                                                                  |
| Full Name                                                            | Nam Nghiep Tran                                                                                                                     |
| Email Address                                                        | namnghiep.tran@adelaide.edu.au                                                                                                      |
| Research profile/staff homepage URL                                  | <a href="https://researchers.adelaide.edu.au/profile/namnghiep.tran">https://researchers.adelaide.edu.au/profile/namnghiep.tran</a> |
| Current University                                                   | UoA                                                                                                                                 |
| Current School/Academic Unit                                         | Chemical Engineering                                                                                                                |
| Is the Principal Supervisor a staff member or title holder/affiliate | Yes                                                                                                                                 |
| Is the Principal Supervisor the primary contact for this project     | Yes                                                                                                                                 |

## Supervisor Panel

|                              |                                                                                                                       |
|------------------------------|-----------------------------------------------------------------------------------------------------------------------|
| Co-Supervisors details:      | Prof. Dr. David Lewis<br>Executive Dean<br>Merger Integration Management Office<br>Division of University Integration |
| External Supervisor details: | TBD                                                                                                                   |
| Any other comments:          | TBD                                                                                                                   |